The matte is separated from the slag and fed to the converting furnace. During conversion, the FeS present in the matte is first oxidised to FeO. This is removed by slag formation with silica. The remaining copper sulphide is further oxidised to its oxide which is subsequently converted to metallic copper as shown below.

$$2Cu_2S\ (l,s) + 3O_2\ (g) \longrightarrow 2Cu_2O\ (l,s) + 2SO_2\ (g)$$

$$2Cu_2O\ (l) + Cu_2S\ (l) \longrightarrow 6Cu\ (l) + SO_2\ (g)$$

The metallic copper is solidified and it has blistered appearance due to evolution of SO_2 gas formed in this process. This copper is called blistered copper.

Reduction by carbon:

In this method the oxide ore of the metal is mixed with coal (coke) and heated strongly in a furnace (usually in a blast furnace). This process can be applied to the metals which do not form carbides with carbon at the reduction temperature.

Examples:

$$ZnO\ (s) + C\ (s) \longrightarrow Zn\ (s) + CO\ (g)\uparrow$$

$$Mn_3O_4\ (s) + 4C\ (s) \longrightarrow 3Mn\ (s) + 4CO\ (g)\uparrow$$

$$Cr_2O_3\ (s) + 3C\ (s) \longrightarrow 2Cr\ (s) + 3CO\ (g)\uparrow$$

Reduction by hydrogen:

This method can be applied to the oxides of the metals (Fe, Pb, Cu) having less electro-positive character than hydrogen.

$$Ag_2O\ (s) + H_2\ (g) \longrightarrow 2Ag\ (s) + H_2O\ (l)$$

$$Fe_3O_4\ (s) + 4H_2\ (g) \longrightarrow 3Fe\ (s) + 4H_2O\ (l)$$

Nickel oxide can be reduced to nickel by using a mixture of hydrogen and carbon monoxide (water gas)

$$2NiO\ (s) + CO\ (g) + H_2\ (g) \longrightarrow 2Ni\ (s) + CO_2\ (g) + H_2O\ (l)$$

Reduction by metal:

Metallic oxides such as Cr_2O_3 can be reduced by an aluminothermic process. In this process, the metal oxide is mixed with aluminium powder and placed in a fire clay crucible. To initiate the reduction process, an ignition mixture (usually magnesium and barium peroxide) is used.

XII U1 Metallurgy - Jerald.indd 9 04-01-2021 12:41:34